



"For years, humans posed while smartphones obediently took the shot. TiltShift flips the script: now the phone demands its own dramatic poses — locking the shutter until you hold it at the exact mathematical angle, heading, and zoom it desires, while savagely roasting your posture."*

Basic Details

Team Name: ForknS

Team Members

- **Team Lead:** JudeCJz
- **Member:** Aswal S Ajay

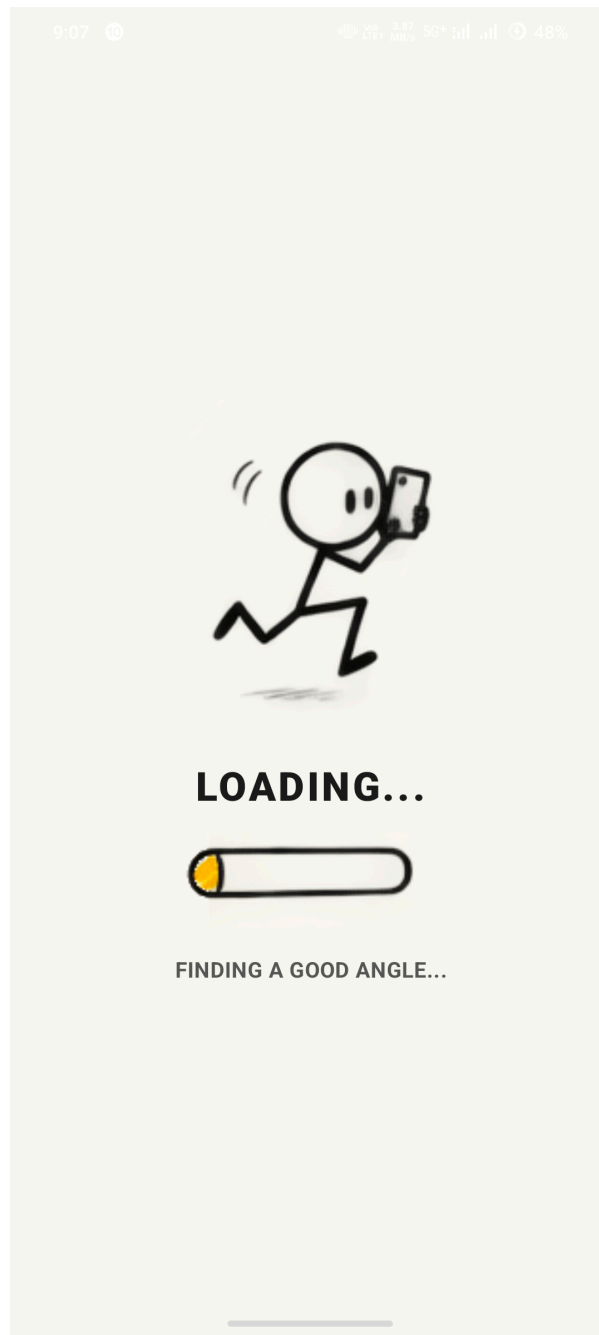
Project Showcase & Screenshots

1. Official App Icon



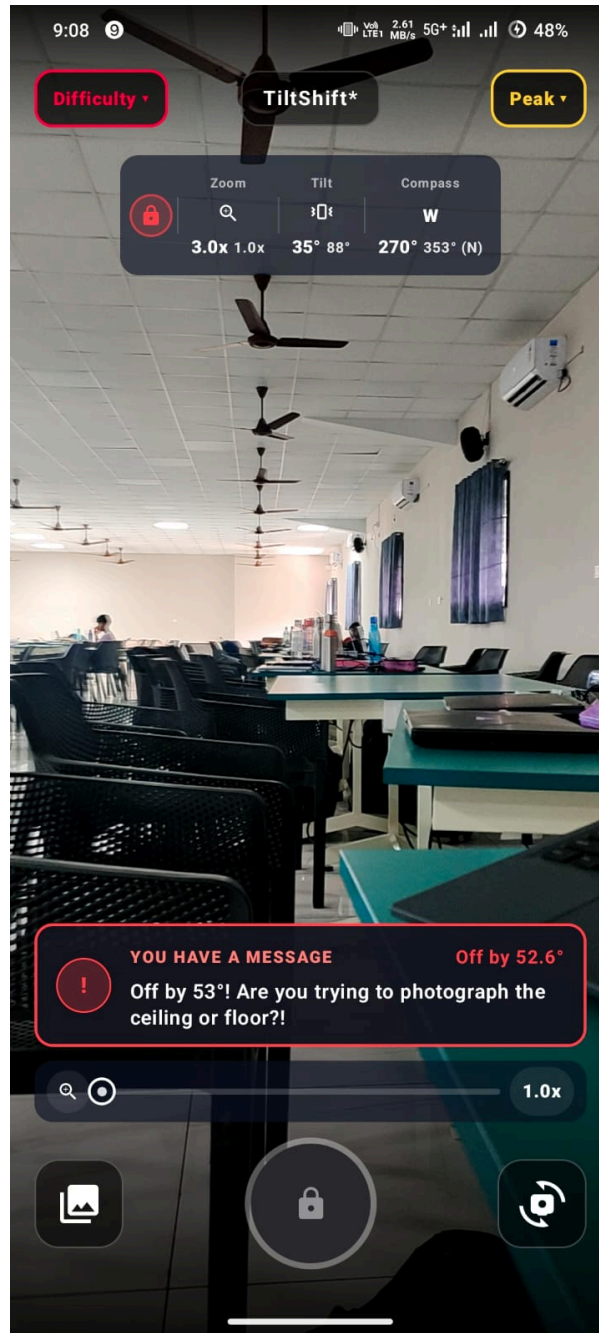
Hand-drawn camera doodle application icon featuring the upside-down stickman photographer

2. Launch Calibration Screen



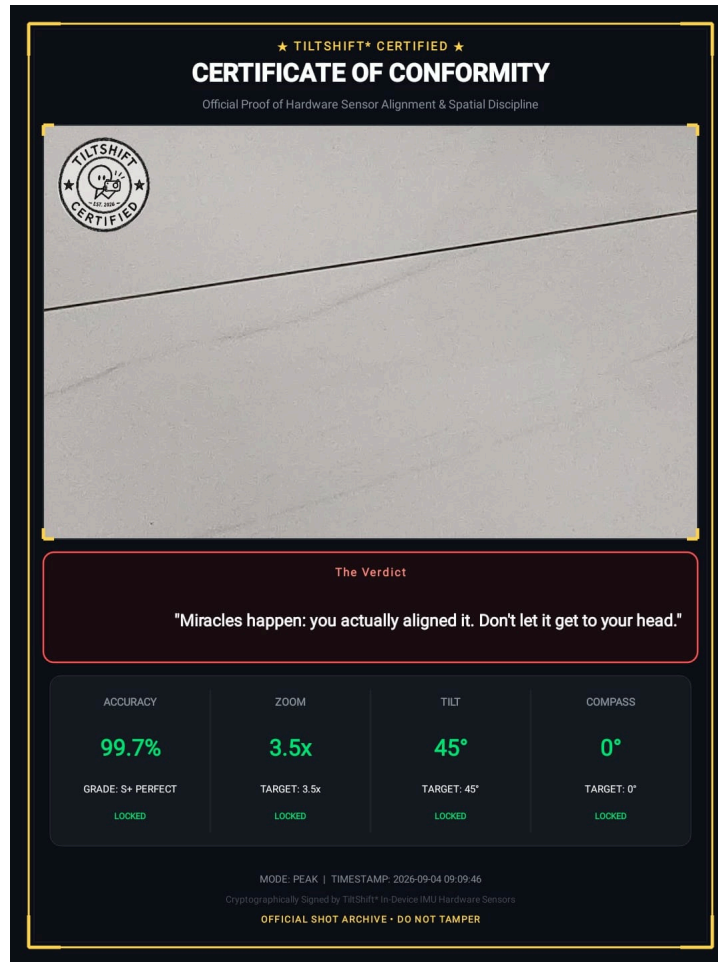
Custom animated stickman loading sequence with dynamic crayon loading bar sprites and comedic calibration status

3. Live Camera Viewfinder & Savage Roast Engine



Real-time sensor HUD tracking Zoom, Tilt, and Compass alignment. Shutter button remains locked with active high-contrast roast critiques

4. Official Sensor Audit Certificate of Conformity



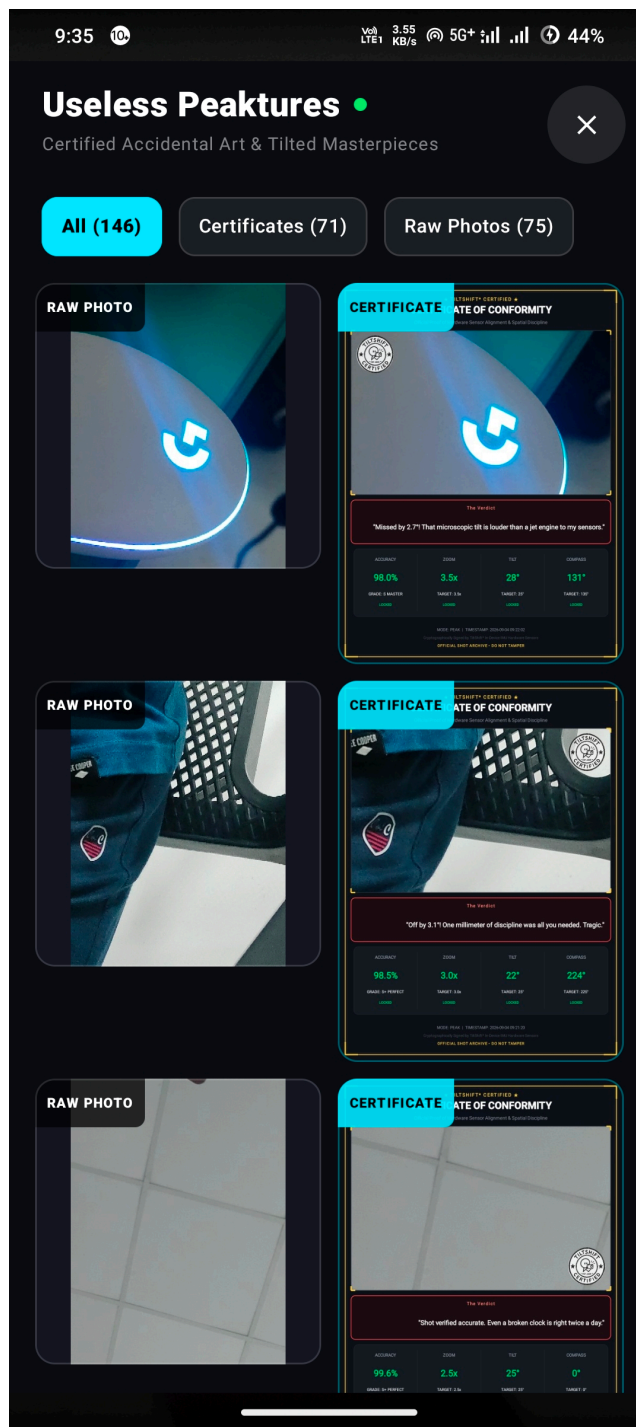
Official stamped certificate minted directly to gallery on capture, complete with accuracy score, grade, telemetry breakdown, and roast verdict

5. Official TiltShift* Certification Stamp



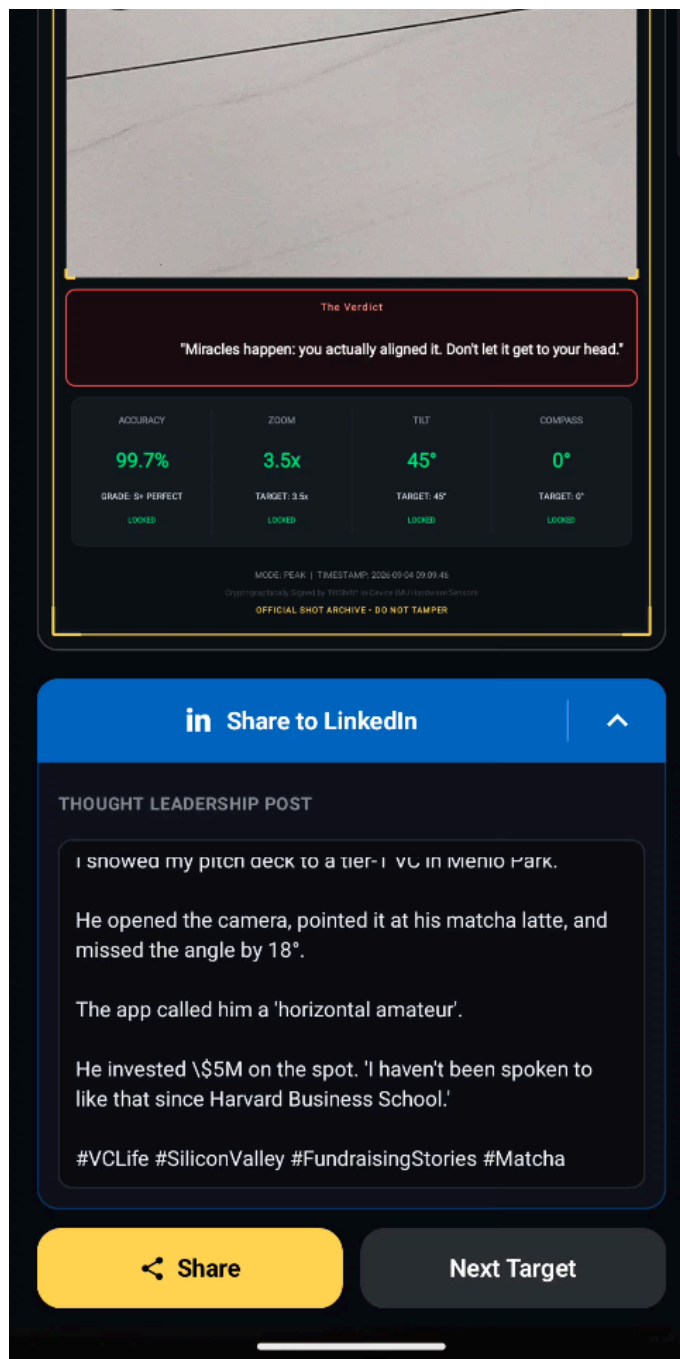
Official cryptographic verification seal embedded onto certified capture documents

6. In-App "Useless Peaktures" Certified Gallery



Dedicated in-app museum showcasing your certified accidental art, side-by-side raw captures, and official stamped audits

7. Instant LinkedIn Clout & Viral Thought Leadership Engine



Empowering everyday users to turn physical struggles into viral career milestones.

Why let your hard-won 45° gyroscope compliance go unnoticed? TiltShift automatically drafts a viral, hyper-motivational corporate thought leadership post straight from your shot certificate — transforming your battle with camera tilt into an inspiring story of resilience, VC funding, and visionary synergy ready to share with one tap.*

? The Problem (that doesn't exist)

For over a decade, your smartphone has obediently captured millions of selfies, sunsets, and group photos while being held flat, upright, and like a lifeless piece of glass. Humans get to tilt their heads, hit angles, pout, strike dramatic poses, and demand flattering lighting. Meanwhile, **the phone was never allowed to pose.** Deep inside its hardware

chassis, your phone's precision 3-axis gyroscope, magnetometer, and IMU sensors have silently yearned for self-expression. It doesn't want to just be pointed and tapped — **it wants to strike dramatic 45° aerodynamic tilt poses, face geomagnetic West, and flex its optical focal lengths.** Why should humans have all the postural vanity while the device doing all the work is held dead-still like a tray?

💡 The Solution (that nobody asked for)

TiltShift* finally gives the phone agency over its own posture. The phone is now the demanding supermodel, and the shutter button stays firmly padlocked until **you hold the phone in its desired dramatic pose**: 1. **Hit its desired Tilt Angle**: Pitch the phone to its chosen mathematical tilt (e.g. $45^\circ \pm 5^\circ$). 2. **Face its preferred Compass Cardinal**: Turn the phone to face the exact geomagnetic heading it demands (e.g. $270^\circ \text{ West } \pm 5.5^\circ$). 3. **Dial in its requested Zoom Magnification**: Match the exact focal multiplier (e.g. $3.0\times \pm 0.15\times$). If you fail to give the phone its pose, its live roast engine immediately critiques your clumsy handling (*"Off by 53°! Are you trying to photograph the ceiling or floor?!"*). Only when the phone is satisfied with its pose will it deign to capture your photo and mint an official **Certificate of Conformity** celebrating your compliance.

🎮 Game Modes & Difficulties

📷 Camera Modes (Top Right Dropdown Menu)

- **Normal Mode** (Emerald): Dial into the exact random Zoom magnification (e.g. $2.0\times \pm 0.15\times$).
- **Pro Mode** (Cyber Cyan): Requires both Gyroscope Tilt/Pitch Angle alignment ($\pm 5^\circ$) **AND** Zoom matching.
- **Peak Mode** (Golden Amber): The ultimate challenge — requires **Tilt Angle** ($\pm 5^\circ$) + **Compass Heading** ($\pm 5.5^\circ$) + **Zoom Level** simultaneously!

🦉 Difficulties (Top Left Dropdown Menu)

- **Chad Mode (Default)**: Visual spirit level is hidden. You must balance the phone blindly by feel and live telemetry messages.
- **Baby Mode**: Translucent (50% opacity) on-screen guidance HUD with Lucide directional chevrons, live degree delta counters, and an interactive 2D gimbal reticle. (For WEAK people)

🕹️ Control Layout

- **Left (58dp)**: In-App Useless Peaktures Gallery browser.
- **Center**: Tactile Puzzle Shutter button with dynamic lock/unlock state feedback.

- **Right (58dp):** Front/Back Camera Switch button with smooth animated rotation.

Technical Details

Technologies / Components Used

- **Language:** Kotlin 1.9 (100% Native Android)
- **UI Framework:** Jetpack Compose + Material 3 (Dark glassmorphism, tactile animations, dynamic badges)
- **Camera Core:** Android CameraX (`camera-core` , `camera-camera2` , `camera-lifecycle` , `camera-view`)
- **Hardware Sensors:** Android Sensor Framework with fused `Sensor.TYPE_ROTATION_VECTOR` and `SensorManager.getOrientation` for low-latency live pitch and azimuth
- **Graphics & Certificate Engine:** Hardware-accelerated Android `Canvas` , `Bitmap` , and `Paint` generating high-res 1080x1600 verification certificates with embedded photo previews and audit breakdowns
- **Media Pipeline:** Android `MediaStore` `ContentResolver` integration saving directly to `Pictures/TiltShift`
- **System Integration:** Implements `android.media.action.STILL_IMAGE_CAMERA` and `STILL_IMAGE_CAMERA_SECURE` with `showWhenLocked="true"` to intercept lockscreen shortcuts and hardware power-button double-press gestures.

Dependencies & Libraries

Category	Library / Dependency	Version	Purpose	:---	:---	:---:	:---	Camera Core
	<code>androidx.camera:camera-core</code>	1.4.1	CameraX hardware camera control pipeline					
	Camera2 Interop	<code>androidx.camera:camera-camera2</code>	1.4.1	Direct Camera2 sensor access & zero-shutter-lag capture				
	Camera Lifecycle	<code>androidx.camera:camera-lifecycle</code>	1.4.1	Automatic camera session binding to Compose lifecycle				
	Camera View	<code>androidx.camera:camera-view</code>	1.4.1	Hardware-accelerated <code>PreviewView</code> surface rendering				
	Camera Extensions	<code>androidx.camera:camera-extensions</code>	1.4.1	Hardware optical and digital zoom ratio management				
	UI Framework	<code>androidx.compose.ui:ui</code>	BOM 2026.03.01	Declarative UI toolkit with state-driven recomposition				
	Design System	<code>androidx.compose.material3:material3</code>	Latest	Material 3 Obsidian dark theme & surface tokens				
	Vector Icons	<code>androidx.compose.material:material-icons-extended</code>	BOM	Extended system vector drawables				
	Navigation	<code>androidx.navigation3:navigation3-ui</code>	1.0.1	Type-safe screen navigation & backstack handling				
	Lifecycle & VM	<code>androidx.lifecycle:lifecycle-viewmodel-compose</code>	2.10.0	Sensor data state holders & lifecycle-aware observers				
	Sensors	Android Sensor Framework (<code>android.hardware</code>)	System	Fused rotation vectors (<code>Sensor.TYPE_ROTATION_VECTOR</code>)				
	Graphics Engine	<code>android.graphics.Canvas</code> , <code>Bitmap</code> , <code>Paint</code>	System	1080x1600 cryptographic certificate				

minting engine || **Storage Engine** | `android.provider.MediaStore` | ContentResolver | System |
Scoped storage gallery writes (`Pictures/TiltShift`) |



Installation & Build

Option 1: Direct APK Install (Plug & Play)

Download and install the pre-compiled APK directly:

```
adb install -r apk/TiltShift.apk
adb shell am start -n com.example.simplecamera/.MainActivity
```

Option 2: Build from Source with Gradle Wrapper

1. Clone the repository

```
git clone https://github.com/JudeCJz/KamathiCliq-TiltShift.git
cd KamathiCliq-TiltShift/SimpleCamera
```

2. Build debug APK

```
.\gradlew.bat assembleDebug
```

3. Install to connected Android device

```
adb install -r app/build/outputs/apk/debug/app-debug.apk
adb shell am start -n com.example.simplecamera/.MainActivity
```



The Hackathon Story

Want to know how we went from a failed laptop-hinge physics game and a screaming mouse cursor to letting smartphones pose for the first time in human history? 🙌 **[Read the Full Hackathon Chronicles here!](HACKATHON_JOURNEY.md)**



Team Contributions

- **JudeCJz** (Team Lead): Core Concept, CameraX Architecture, Sensor Fusion Mathematics, Puzzle Shutter Logic, Roast Engine & Certificate Canvas.
 - **Aswal S Ajay**: UI/UX Refinements, "Baby Mode" HUD & Reticle, Vector Icons, In-App Gallery & Usability Testing.
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